Lab 7 APS reference specifications

Use these specifications with the APS use case diagram in Lab 6, Figure 2. UC001 follows the published worked example; UC003 and UC006 supply solutions to its two tasks. Keep their IDs, system boundary and outcomes when deriving the Lab 7 class diagram.

Use case ID:	**UC001 Register client**
Brief description	The client creates a new registration with APS and PoB is informed about the new client.
Primary actors	Client.
Secondary (supporting) actors	PoB.
Trigger	The client requests registration by providing a name and address.
Preconditions	Any required login checks have been completed outside APS.
Postconditions	A registration with a unique number exists. PoB has been notified of that number and has opened the corresponding financial account.
Main Success Scenario	
Actor Action	**System Response**
1. The client provides name and address.	2. APS checks whether the client already has a registration (see 2.a). 3. APS creates the registration with the client's details and assigns a unique registration number. 4. APS sends the assigned registration number to PoB, which opens the client's financial account under that number. 5. APS informs the client of the registration number.
Alternative flows: 2.a. Existing client: if the client already exists, APS informs the client that a new registration cannot be created. The use case ends without creating another registration.	

Use case ID:	UC003 List item for sale
Brief description	A registered client lists a phone for sale by auction or at a fixed price. APS records the item, prepares a display image for the seller's approval and includes the approved item in the product catalog.
Primary actors	Client, acting as seller.
Secondary (supporting) actors	None.
Trigger	The client chooses to sell an item.
Preconditions	Login has been checked outside APS. The client has a registration.
Postconditions	The item is recorded with its ID, details, sale type and terms, attached to the client's registration, and displayed in the product catalog with the approved image.
Main Success Scenario	
Actor Action	**System Response**
1. The client selects the type of item: smartphone, normal mobile phone or land phone (see 1.a).	
2. The client provides the model, make, capacity and colour.	
3. The client selects sale by auction and gives the starting price and the auction end time (see 3.a).	4. APS assigns an ID to the item. 5. APS records the item information and sale terms and attaches them to the client's registration. 6. APS creates a display image from the item description and asks the client to approve it.
7. The client approves the display image (see 7.a).	8. APS includes the item in the product catalog and displays it for sale.
Alternative flows: 1.a. Smartphone: the client also selects the sub-category, Apple iPhone or Samsung Galaxy. The use case resumes at step 2. 3.a. Fixed-price sale: the client selects fixed-price sale and gives the price. The use case resumes at step 4. 7.a. Image not approved: APS re-creates the display image and asks the client again. The use case resumes at step 7.	

Use case ID:	UC006 Place bid
Brief description	A registered client offers a price for an item on auction. APS accepts the bid if the auction is still open and the bid is greater than the last offered bid, and records it in the client's registration.
Primary actors	Client, acting as buyer.
Secondary (supporting) actors	None.
Trigger	The client selects an item on auction to bid on.
Preconditions	Login has been checked outside APS. The client has a registration. The item is displayed in the catalog for auction.
Postconditions	The bid is recorded in the client's registration and is the last offered bid for the item.
Main Success Scenario	
Actor Action	**System Response**
1. The client selects an item on auction.	2. APS checks that the auction end time has not passed and shows the last offered bid (see 2.a).
3. The client submits a bid price.	4. APS checks that the auction end time has not passed and that the bid is greater than the last offered bid (see 4.a and 4.b). 5. APS records the bid in the client's registration as the last offered bid. 6. APS confirms the accepted bid to the client.
Alternative flows: 2.a. Auction ended: APS informs the client that the auction has ended. No bid is recorded and the use case ends. 4.a. Auction ended before the bid was submitted: as 2.a. 4.b. Bid not greater than the last offered bid: APS rejects the bid and informs the client. The use case resumes at step 3.	